

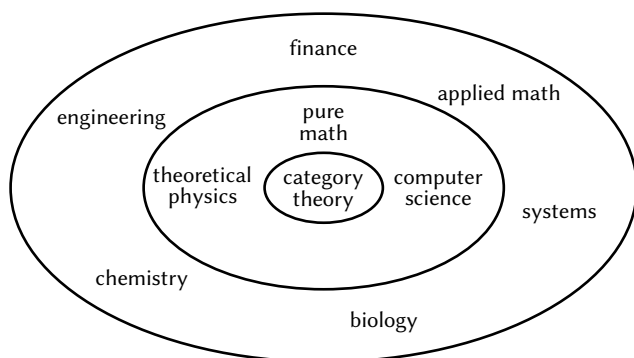
stay at home. I actually once spoke to a representative from a health insurance company who thought this was the case, and did not comprehend the concept that I might visit a different state and ask about coverage there.

One particular feature of this book is that I will not demand that the reader does any exercises in order to follow the book. It is standard in math books to exhort the reader to work through exercises, but I believe this is offputting to many non-mathematicians, as well as some mathematicians (including me). I will provide “Things to think about” from time to time, but these will really be questions to ponder rather than exercises of any sort. And one of the main purposes of those questions will be to develop our instincts for the sorts of questions that mathematicians ask. The hope is that as we progress, the reader will think of those questions spontaneously, before I have made them explicit. Thinking of “natural” next questions is one important aspect of mathematical thinking. Where working through them is beneficial to understanding what follows I will include that discussion afterwards.

The content in this book

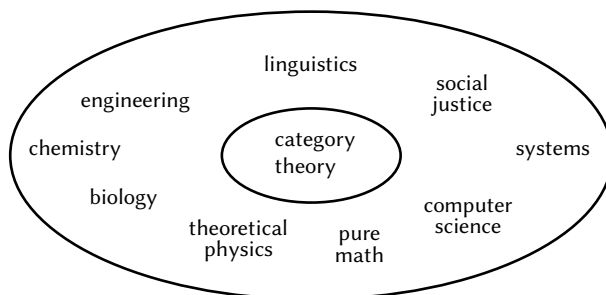
Category theory was introduced by Eilenberg and Mac Lane in the 1940s and has since become more or less ubiquitous in pure mathematics. In some fields it is at the level of a language, in others it is a framework, in others a tool, in others it is the foundations, in others it is what the whole structure depends on.

Category theory quickly found uses beyond pure math, in theoretical physics and computer science. The view of things at the end of the 20th century might be regarded like this, with the diagram showing applications moving outwards from category theory:



However, since then category theory has become increasingly pervasive,

finding direct applications in a much wider range of subjects further from pure mathematics, such as ecological diversity, chemistry, systems control, engineering, air traffic control, linguistics, social justice. The picture now might be thought of as more like this:



For some time the only textbook on the subject, from which everyone had to try and learn, was the classic graduate text by Mac Lane, *Categories for the Working Mathematician* (from 1971). The situation was this: there was a huge step up to Mac Lane, which many people, even those highly motivated, failed to be able to reach.

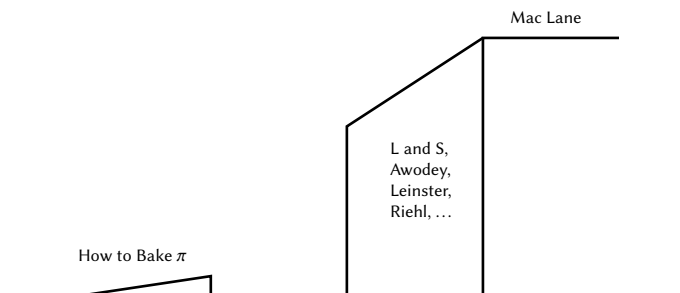
Mac Lane

As is the way with these things, what started as a research field had become something that graduate students (tried to) study, and eventually it trickled down into a few undergraduate courses at individual universities around the world that happened to have an expert there who wanted to teach it at that level. This spawned several much more approachable textbooks at the turn of the 21st century, notably those by Lawvere and Schanuel (1997), followed by a sort of second wave with Awodey (2006), Leinster (2014) and Riehl (2016). There was still a gap up to those books, and the gap was still insurmountable for many people who didn't have the background of an undergraduate mathematician, either in terms of experience with the formality of mathematics or background for the examples being used.[†]

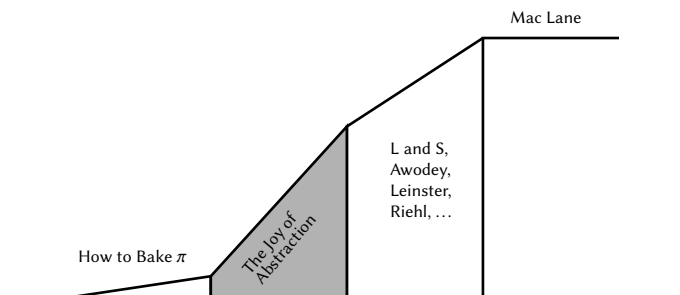
In 2015 I wrote *How to Bake π* , a book about category theory for an entirely

[†] Lawvere and Schanuel include high school students in their stated target audience but I think they have in mind quite advanced ones. There are also some recent books aimed at specific types of audience, which are less in the vein of standard textbooks; see Further reading.

general audience with no math background whatever. The situation became like this:



How to Bake π provides a ramp from more or less nothing but does not get very far into the subject and remains mostly informal throughout. The role of the present work is to fill the remaining gap:



This book will not assume undergraduate level mathematics training, nor even a disposition for it, but it will become technical. The aim, after all, is to bridge the gap up to the undergraduate textbooks. We will build up very gradually towards the rigorous formality of category theory. While there is no particular technical prerequisite for learning category theory, the formality of any mathematical field can be offputting to those who are not used to it. The intention is that you can read this book *and turn the pages*, in contrast with so many math books where you have to sit and think for an hour, week or month about a single paragraph (although those books have their place too). It will be informal and personal in style, including descriptions of my personal feelings about various aspects of the subject. This is not strictly part of mathematics but I believe that building an emotional engagement is an important part of learning anything. There will also be many diagrams to help with visualizing, partly to engage visual people and partly because the subject is very visual. There will be both formal mathematical diagrams, and informal “schematic” diagrams. All these things mean that the book will in some sense be the op-